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Thermosonic Technology: Revolutionizing Heavy Oil Flow Assurance and Production Optimization

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Abstract

Heavy oil operators around the world often experience production challenges during colder periods due to increased oil viscosity. In Oman's Asset-M, winter-related shutdowns posed recurring challenges, which were mitigated with the introduction of Thermosonic technology offering a chemical-free, low-energy approach. Elevated flowline backpressure is mainly caused by heavy oil viscosity, low GOR, low water-cut, and fluctuating ambient temperatures [1-2]. Thermosonic technology was applied to liquefy the heavy oil and enhance flow mobility, enabling sustained production and unlocking additional potential in these wells. This paper will present the technology application for the first time and show case real case studies from Oman.

Thermosonic technology employs electrophysical effects to convert electrical energy into targeted thermal and acoustic energy. Its dual-action mechanism combines localized induction heating (via high-frequency electromagnetic fields) and acoustic stress wave pulsation, reducing oil viscosity and surface adhesion. It is applied through a specialized cable along surface flowlines generated by specialized units. This approach enhances fluid mobility with significantly lower energy consumption than traditional methods like steam injection or resistive heating. Strategically installed Thermosonic units ensure continuous, safe, and efficient treatment of viscous oil across flowlines, optimizing performance without operational disruptions [3-4].

The Thermosonic technology was deployed in 4 wells in PDO south assets to demonstrate technology capability. Thermosonic technology successfully restored and optimized production in 4 targeted heavy oil wells, demonstrating clear effectiveness in mitigating flowline backpressure. Real-time monitoring confirmed a measurable reduction in backpressure, while external pipeline temperatures increased to mimic optimal summer conditions—all within safe thermal limits (<65°C). These improvements enabled the non-producing wells (impacted by cold weather conditions and elevated oil viscosity) to resume continuous production year-round and minimizing deferment, with several wells exceeding historical production rates.

Beyond reviving non-producing heavy oil wells, the technology's low-energy & non-chemical approach proved effectiveness compared to conventional methods (e.g., heating cables or steam injection), which were either inefficient, energy-intensive or economically unviable.

The results confirm the potential of Thermosonic technology to unlock marginal reserves, extending well life, and improving recovery rates in challenging heavy oil reservoirs offering: restoration of deferred production, improved flow assurance without chemical dependency and lower energy consumption compared to traditional thermal methods.

Thermosonic is a novel, physics-driven technology that eliminates the need for chemical agents or additives in managing viscous oil flow challenges. Its non-intrusive design extends equipment run-life, reduces maintenance cycles, and operates with lower energy, making it both economically and environmentally superior in comparison to conventional flow assurance techniques.

This paper presents field-proven results from PDO's southern assets, where Thermosonic:

- Restored 100% of deferred production in Well M-4 (previously non-flowing in winter).
- Reduced flowline backpressure by 20–45%, enabling continuous operation.
- Increased Well S-3's production by 35% post-installation.
- Increased Flowline temperature up to $<65^{\circ}\text{C}$, protecting plastic-lined flowlines while mimicking optimal summer conditions.

With 50% lower energy use vs. steam injection and zero chemical dependency, Thermosonic offers a scalable, retrofit-ready solution for heavy oil fields.

Introduction

A significant portion of the Sultanate of Oman's oil and gas production originates from its southern assets, which are widely scattered and predominantly produce heavy oil. At the heart of these southern fields lies Asset-M, a key contributor in Petroleum Development Oman production portfolio. Most of the fields in this region are developed under a waterflood strategy, with production mainly characterized by high-viscosity crude oil.

One of the critical operational challenges encountered in these heavy oil fields is the severe impact of viscosity, particularly during the early stages of well life when wells are producing at low water cuts. This challenge is exacerbated during the winter months, when ambient temperatures can drop to as low as $5\text{--}10^{\circ}\text{C}$, especially during the early morning hours. The reduced temperature significantly increases the oil viscosity, impeding flow within the surface flowlines and elevating backpressure. Once the flowline pressure reaches its maximum threshold, the safeguarding systems trigger a well trip to protect surface equipment, resulting in considerable production deferment and value loss.

This viscosity-related flow assurance issue, while temporary, demands an innovative yet simple and flexible solution that can be deployed during the early production phase—until the water cut increases and the challenge naturally subsides. In this context, Thermosonic Technology has emerged as a novel approach, combining the benefits of thermal and acoustic energy to effectively reduce oil viscosity and restore production rates.

This paper presents the successful deployment of Thermosonic Technology in one of Asset-M's high-viscosity oil fields. The case study demonstrates how this intervention has mitigated winter production deferment, improved flow assurance, and ultimately enhanced value realization from wells that would otherwise remain constrained.

Overview of Asset-M and the need for a novel solution

Asset-M holds historical significance as one of the earliest developed oil and gas assets in Oman producing from shallow, unconsolidated sandstone formations. The asset comprises more than ten actively developed reservoirs, all producing predominantly heavy and viscous crude oil (Figure-1). These fields exhibit relatively good porosity, typically ranging between 15% and 25%, although permeability varies significantly—from as low as 10 millidarcies (md) in some reservoirs to over 1000 md in others, especially in the more productive zones. The produced oil is characterized by low gas-oil ratios (GOR), ranging from 1 to 24 m³/m³, and the presence of sour components such as H₂S and CO₂ is minimal, contributing to a relatively benign production environment. All reservoirs are under waterflood development to maintain reservoir pressure, yet they remain under sub-hydrostatic conditions, which further simplifies reservoir management. Given these geological and operational characteristics, Asset-M is generally considered a low-risk asset. However, one of the key challenges stems from the high viscosity of the crude, which ranges from 5 to 1000 centipoise (cp) at reservoir temperature and API gravities between 18 and 25. During winter, when ambient surface temperatures drop, oil viscosity at the surface can exceed 5000 cp, leading to significant flow assurance issues that impact production performance and operational reliability.

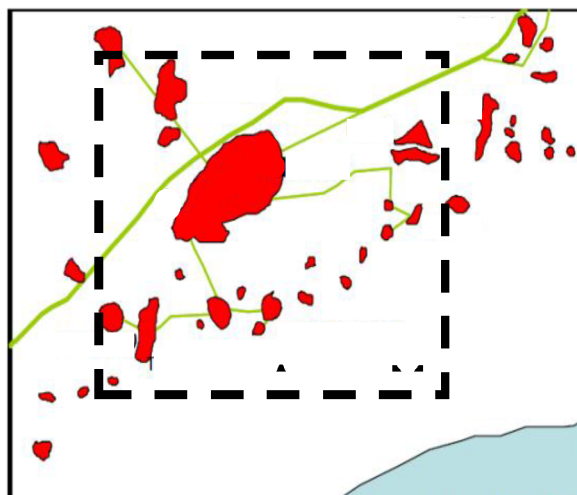


Figure 1—The diagram is showing a map of Asset-M and the scattered fields around it. Most of the high oil viscosity fields are remote and slightly far from the production facility.

Wells producing high-viscosity crude oil in Asset-M are particularly vulnerable to flow assurance challenges during the winter season, especially in the early morning hours when ambient surface temperatures can drop to as low as 5–10°C. This temperature decline significantly increases oil viscosity at the surface, resulting in restricted fluid mobility through production flowlines and a corresponding rise in backpressure. Once the flowline pressure approaches or exceeds its design threshold, the well's safeguarding system is triggered, causing an automatic shutdown to protect surface equipment from overpressure-related failures. These repeated shutdowns, common during the colder months, contribute to substantial production deferment and associated revenue loss.

An illustrative case is presented in Figure-2, which shows a well equipped with a Progressive Cavity Pump (PCP) experiencing this issue. The PCP rod speed trend, indicated by the green-colored line, clearly demonstrates frequent tripping events during winter, in contrast to the stable and uninterrupted operation observed during the warmer summer period. Similarly, the back pressure, represented by the orange-colored line, shows that the well trips once the pressure reaches the safeguarding system's trigger threshold. This operational behavior highlights the direct impact of temperature-driven viscosity changes on well performance and emphasizes the need for effective mitigation strategies to maintain production continuity.

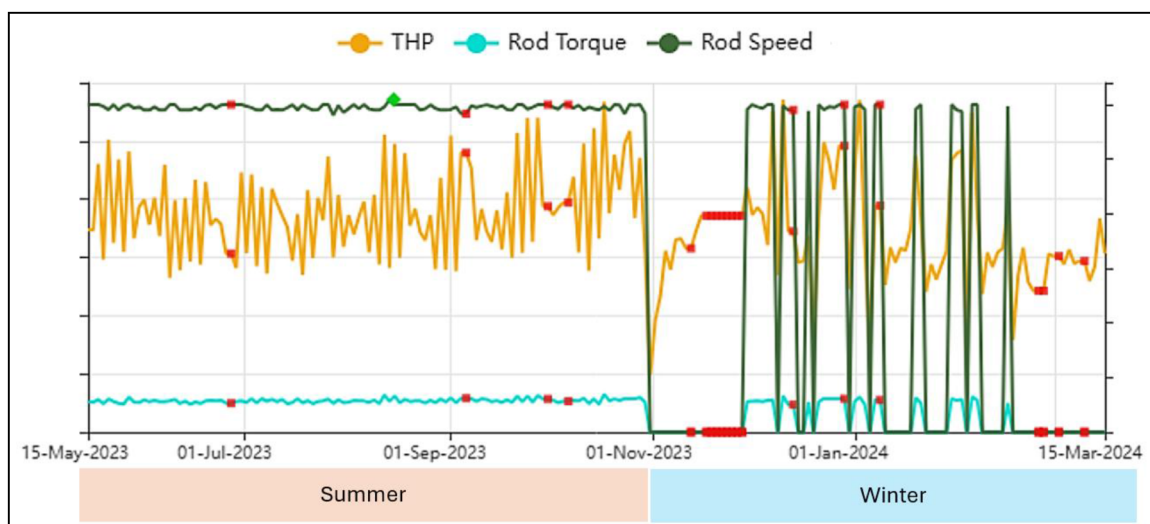


Figure 2—Plot showing an example of a trend of a well tripping during winter and continuously producing during summer.

Since this challenge is temporary and typically subsides as water cut increases, the proposed solution must be simple, effective, innovative, and flexible enough to be relocated and reused in other affected wells. Conventional heaters were ruled out as an option because the flowlines are internally lined with plastic, and the temperature must not exceed 65°C to avoid damaging the liner. Thermosonic Technology, by contrast, employs electrophysical principles to convert electrical energy into targeted thermal and acoustic energy—while maintaining safe operating temperatures below 65°C—making it a suitable and non-intrusive solution for this challenge.

Thermosonic Technology

Heavy oil production continues to pose significant operational challenges due to its naturally high viscosity, low gas-oil ratios (GOR), and pronounced sensitivity to ambient temperature fluctuations. Conventional mitigation strategies—such as steam injection and resistive heating cables—often demand substantial energy input, complex infrastructure, and may prove uneconomical for marginal or low-rate wells. In contrast, Thermosonic technology introduces a novel, physics-based solution that integrates electromagnetic induction heating with acoustic pulsation effects. This dual-action approach enables effective viscosity reduction and enhances fluid mobility along surface flowlines, all without relying on chemical additives or excessive energy consumption (Figure-3). The technology proves efficiency in its applications: it prevents paraffin, wax, hydrate, and asphaltene blockages in wells and flowlines; reduces heavy oil viscosity and associated friction losses; reactivates suspended wells affected by flow restrictions; and ultimately boosts the productivity of heavy oil wells.

The Thermosonic system operates using a specialized TS cables strategically distributed along targeted surface flowlines and powered using the standard power supply available at the well location. It harnesses two synergistic physical mechanisms—induction heating and acoustic pulsation—to enhance the flow of heavy crude oil. The induction heating mechanism involves the generation of a high-frequency electromagnetic field (typically 20–60 kHz) by the power unit, transmitted through the TS cables. This field induces eddy currents in the adjacent conductive metal of the steel flowline, producing localized Joule heating. This uniform and controlled heating raises the temperature along the pipe wall, effectively lowering oil viscosity while maintaining safe external temperatures (below 65°C), thus improving fluid mobility. In parallel, the system delivers low-intensity, high-frequency acoustic waves (1100–2000 Hz) that travel along the pipeline wall. These mechanical vibrations generate micro-pulsations that disturb the adhesion of heavy molecular components—such as waxes, paraffins, and asphaltenes—on the inner pipe

surface. By disrupting the boundary layer and reducing drag through surface shear modulation, the acoustic effect enhances oil flow and complements the thermal action. Unlike conventional thermal or chemical treatment methods, Thermosonic requires no injection of external fluids or reagents, is minimally invasive to existing infrastructure, and can be easily retrofitted on current surface lines. It is especially effective in challenging environments characterized by low water cut (BSW), low GOR, and high viscosity, where traditional approaches are often ineffective or prohibitively expensive.

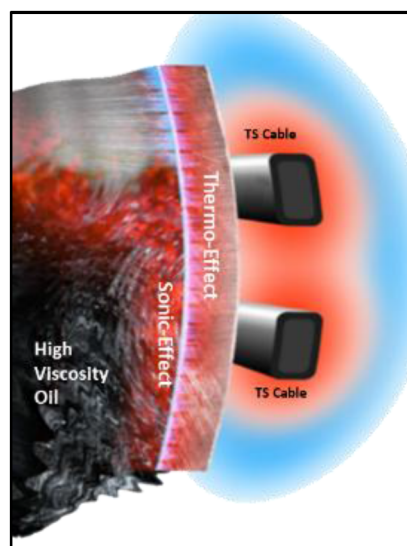


Figure 3—Diagram is showing Thermosonic dual action of thermal energy and acoustic energy.

Thermosonic Candidate Selection and Technology Implementation

Candidate wells for Thermosonic deployment were selected through a structured screening process based on clearly defined production and operational criteria to ensure maximum impact. Priority was given to wells experiencing high deferment due to elevated flowline backpressure, producing heavy crude with an API gravity below 20°, and operating under low gas-oil ratios ($\text{GOR} < 23 \text{ m}^3/\text{m}^3$) and low water cuts ($< 65\%$). Wells with a history of frequent winter-related trips, unstable production during colder periods, or those constrained to minimum operating rates during winter to avoid shutdowns were also identified as strong candidates. For the initial field trial, three wells were selected, each representing a different challenge profile. Table-1 summarizes the viscosity and production data for these wells under summer operating conditions.

Table 1—Viscosity and production data of the candidates selected for Thermosonic technology trail.

Well	Viscosity (cp) at different temperatures	Flow rate (m ³ /d), WC%
S-3	1,593 cp @ 25 °C 662.7 cp @ 40 °C 506.8 cp @ 50 °C	22, 38%
S-4	4,362 cp @ 25 °C 1,365 cp @ 40 °C 707 cp @ 50 °C	37, 61%
M-4	5,039.3 cp @ 25 °C 1,550.1 cp @ 40 °C 969.1 cp @ 50 °C	24, 30%

For this initial pilot, three wells were carefully selected—each representing a distinct challenge scenario relevant to heavy oil operations. Table-1 presents the viscosity characteristics and production data for the

selected wells under typical summer conditions, providing a baseline for performance comparison. The selection aimed to test the Thermosonic system across a range of operational challenges, with three primary objectives in focus:

1. To evaluate the technology's capability to enhance production from wells constrained by elevated viscosity and associated backpressure.
2. To verify its effectiveness in sustaining stable production from wells that frequently shut-in during winter due to poor flow assurance.
3. To assess its potential in reactivating wells that are otherwise incapable of flowing at all during cold weather conditions.

By addressing these three distinct operational challenges, the pilot aimed to validate the versatility, reliability, and value of Thermosonic technology.

Each Thermosonic installation consists of three main components (Figure 4):

1. A central Power Control Unit (PCU) that transforms standard electrical input into a controlled high-frequency output.
2. Custom-engineered Thermosonic (TS) cables built to withstand elevated temperatures, electromagnetic stress, and harsh field conditions. The TS cables are externally clamped along the full length of the surface flowline—from the wellhead to the gathering header—requiring no internal pipeline modifications or rerouting, thus enabling a low-impact, field-friendly installation.
3. Distributed sensor system incorporating temperature sensors and inline pressure transmitters for real-time monitoring and operational control.

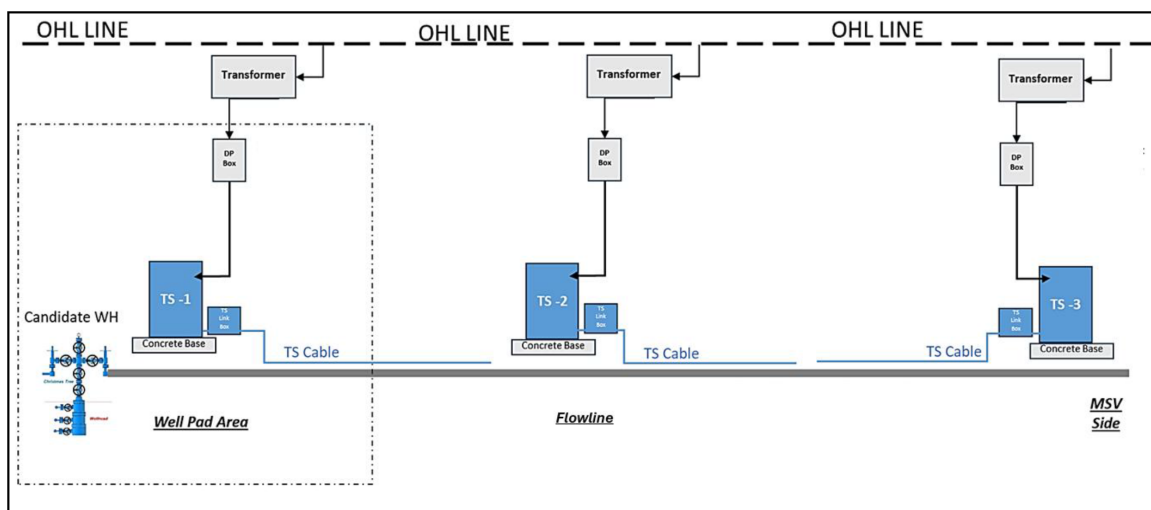


Figure 4—Schematic describing the Thermosonic system component and the installation method.

Results and Discussion

Thermosonic technology was successfully deployed in three heavy oil wells within PDO's southern assets, specifically targeting surface flowlines that were prone to frequent deferments and reduced production potential due to elevated backpressure—particularly during colder months. After the installation and commissioning of Thermosonic units along these flowline segments, real-time operational data showed a marked reduction in flowline backpressure, with pressure drops ranging from 20% to 45% compared to baseline values. This reduction directly contributed to improved flow stability and enhanced mobility of the heavy crude. Additionally, external temperature measurements along the flowlines consistently rose and stabilized between 55°C and 60°C, effectively replicating ideal summer operating conditions even during

winter. These elevated temperatures remained safely below the 65°C threshold, ensuring protection of the internal plastic lining while optimizing crude flow performance [6].

The primary objective behind selecting well S-3 as the first candidate for Thermosonic deployment was to evaluate the technology's effectiveness in enabling well optimization during winter conditions. As shown in Table-1, S-3 has the lowest crude oil viscosity among the group and a relatively low water cut, allowing it to maintain stable flow even in colder months. However, due to increased viscosity at low ambient temperatures, the well could only be operated at low RPM, as illustrated in the figure below (Figure -5). Prior to the Thermosonic installation, multiple attempts were made to bean up the well during winter, but each effort resulted in a trip due to elevated backpressure (indicated with red in Figure-5). Following the installation of the Thermosonic unit, the well was successfully beamed up without any issues, confirming the system's positive impact on reducing backpressure and improving operational flexibility in low-temperature environments. The bean up resulted in an increase in oil production rate by 35% [6].

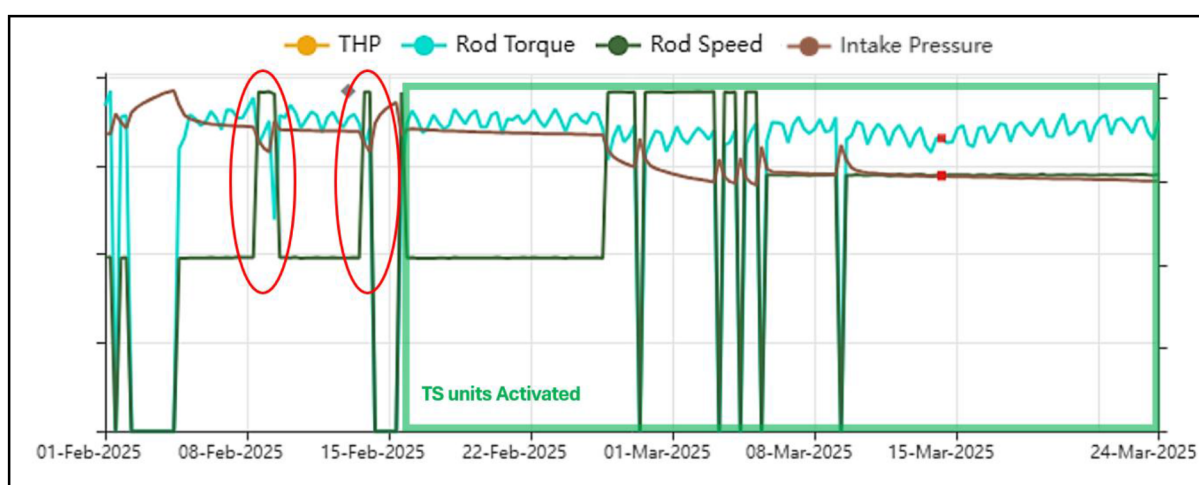


Figure 5—Plot showing the performance of S-3 well before and after installation of Thermosonic unit.

The second candidate selected for Thermosonic deployment, well S-4, was characterized by relatively high crude oil viscosity and a moderate water cut. The key objective in choosing this well was to assess the effectiveness of Thermosonic technology in stabilizing flow in a high-viscosity environment, particularly after an increase in water cut. As illustrated in Figure-6, the well had consistently struggled to sustain stable production during the winter months, frequently experiencing flow interruptions. However, following the installation of the Thermosonic system, a noticeable improvement was observed—S-4 was able to flow smoothly and continuously throughout the remainder of the winter season, confirming the technology's ability to enhance flow stability under challenging conditions [6].

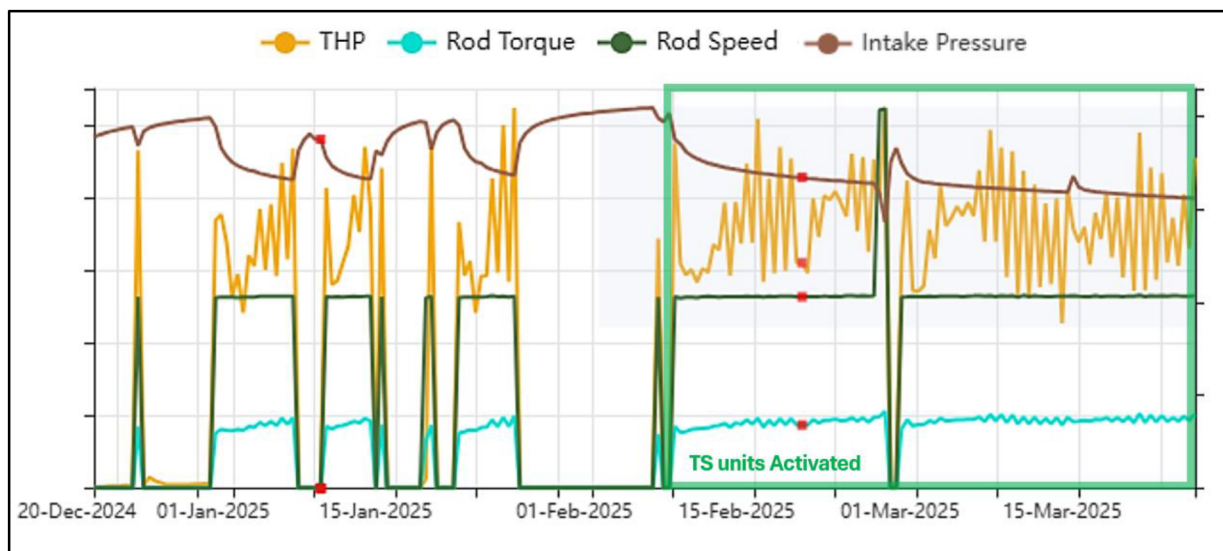


Figure 6—Plot showing the performance of S-3 well before and after installation of Thermosonic unit.

The final candidate selected for Thermosonic deployment was well M-4, which presented a more challenging profile with higher crude oil viscosity and lower water cut compared to S-4. This well was chosen specifically to test the operational limits of the Thermosonic technology under more severe flow assurance conditions. As shown in Figure-7, M-4 was completely unable to flow during the winter period, despite multiple restart attempts—all of which failed due to extreme viscosity and backpressure issues. However, following the installation of the Thermosonic system, the well successfully resumed flow and sustained production throughout the remainder of the winter season. This outcome highlights the robustness and effectiveness of the technology in enabling flow even under the most restrictive and high-viscosity conditions [6].

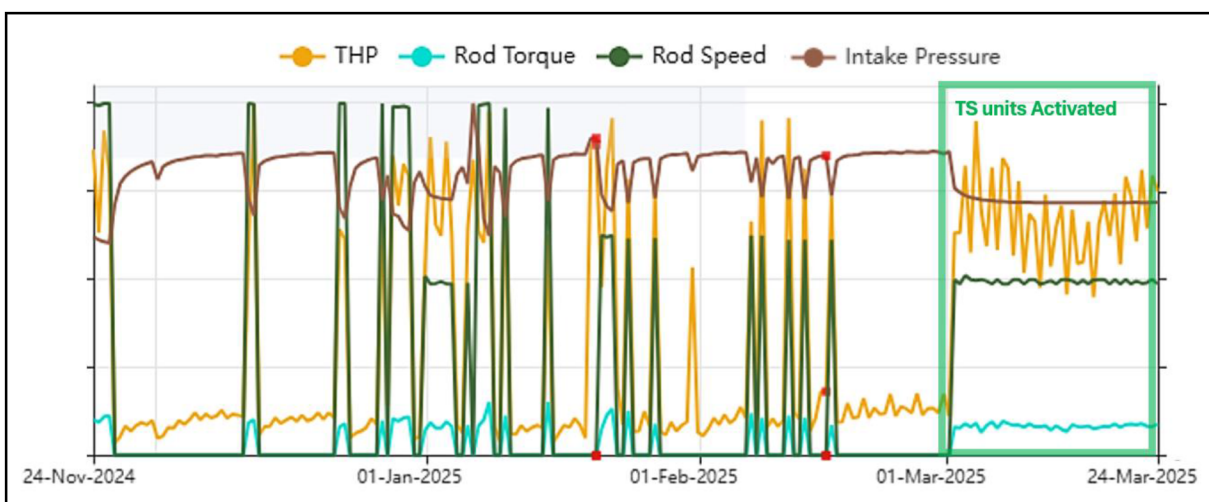


Figure 7—Plot showing the performance of M-4 well before and after installation of Thermosonic unit.

Conclusion

Heavy oil represents a significant portion of global hydrocarbon resources (~70%) [5], but cold-weather conditions continue to challenge production across regions from Canada to Oman. In PDO's Asset-M, winter temperatures ranging from 5–10°C lead to increased oil viscosity, often resulting in well trips and notable

production impacts (~5-10%). Conventional mitigation methods—such as steam injection or resistive heating—can be costly, energy-intensive, and may require substantial surface infrastructure modifications.

Thermosonic technology introduces a dual-action system that combines thermal and acoustic mechanisms to address flow assurance challenges in heavy oil wells. It liquefies heavy oil using targeted induction heating in the range of 20–60 kHz, while simultaneously disrupting wax and asphaltene adhesion through high-frequency acoustic waves between 1,100 and 2,000 Hz. This approach operates with 30–50% less energy compared to conventional heating cables and requires no modifications to the well infrastructure.

The successful deployment of Thermosonic technology in PDO's southern heavy oil fields has demonstrated a step-change in managing flow assurance challenges associated with high-viscosity, low-GOR crude—particularly during the colder months when temperature-driven viscosity spikes typically lead to production instability and deferment. The technology's unique dual-action mechanism—combining localized induction heating and acoustic pulsation—enabled a significant reduction in oil viscosity and surface backpressure without the need for chemical additives, steam injection, or invasive infrastructure upgrades. This physics-based approach provided safe, uniform heating along flowlines while preserving the integrity of plastic linings by maintaining temperatures below 65°C.

Real-time field data collected during the pilot phase confirmed a measurable and consistent drop in backpressure by 20–45%, restoration of flow in previously non-producing wells, and sustained oil production during winter months. Wells S-3, S-4, and M-4 each presented progressively more complex flow challenges, ranging from mild viscosity effects to complete winter shut-ins. In all cases, Thermosonic intervention resulted in notable operational improvements, including a 35% increase in production rate for S-3 and the revival of M-4, which had previously been unable to flow at all during winter. These outcomes affirm the system's ability to perform reliably under varied well conditions, showcasing both technical robustness and adaptability.

Beyond restoring deferred volumes, Thermosonic's low-energy, chemical-free, and non-intrusive nature makes it an economically and environmentally attractive solution. Its quick deployment, real-time monitoring capabilities, and reusability further enhance its field value. As PDO continues to mature and optimize its heavy oil portfolio, especially in assets like Asset-M where viscosity remains a recurring constraint, Thermosonic technology offers a scalable and efficient tool to enhance recovery, extend well life, and reduce production losses associated with seasonal flow assurance issues. With further application and data-driven optimization, this technology holds strong potential to become a key enabler in PDO's broader heavy oil strategy.

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